Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The dimension $[M\,L^{-1}\;T^{-2}]$ corresponds to,

A. moment of force.

B. surface tension.

C. modulus of elasticity.

D. coefficient of viscosity.

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A student measures the time period of 20 oscillation of a simple pendulum five times. That data set is 53 s, 52 s, 55 s, 54 s and 51 s. If the minimum division in the measuring clock is 1 s, then the reported time should be.....

A. 50 ± 1sec

B. 52 ± 2sec

C. 53 ± 2sec

D. 53 ± 3sec

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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A bus moving on a level road with a velocity v can be stopped at a distance of x, by the application of a retarding force F. The load on the bus is increased by 25% by boarding the passengers. Now, if the bus is moving with the same speed and if the same retarding force is applied, the distance travelled by the bus before it stops is:

A. $1.25x$

B. x

C. $5x$

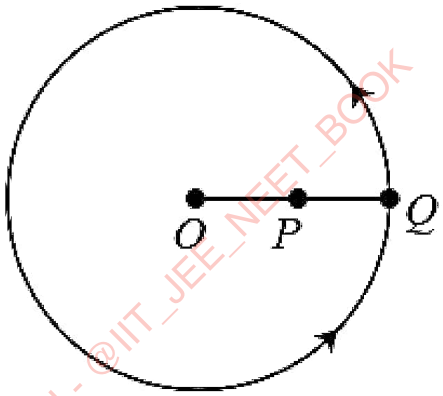
D. $2.5x$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two spheres P and Q, each of mass 200 g are attached to a string of length one metre as shown in the figure. The string and the spheres are then whirled in a horizontal circle about O at a constant angular speed. The ratio of the tension in the string between P and Q to that of between P and O is (P is at mid-point of the line joining O and Q)



A. $\frac{1}{2}$

B. $\frac{2}{3}$

C. $\frac{3}{2}$

D. $\frac{2}{1}$

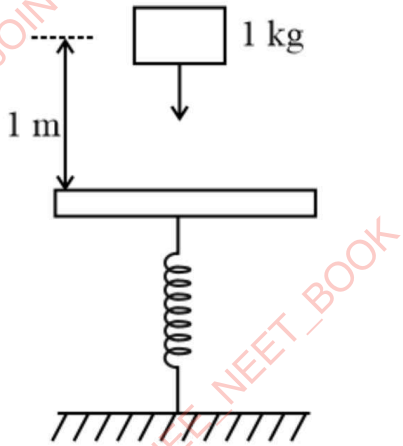
Question No. 5

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Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A mass of 1 kg falls from a height of 1 m and lands on a mass less platform supported by a spring having spring constant 15 N m^{-1} as shown in the figure. The maximum compression of the spring is.
(acceleration due to gravity = 10 m s^{-2})



A. 2 m

B. $\sqrt{2}$ m

C. $(\frac{2}{\sqrt{3}})$m

D. $\sqrt{3}$ m

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two bodies of masses m and $9m$ are placed at a distance r. The gravitational potential at a point on the line joining them, where gravitational field is zero, is (G is universal gravitational constant).....

A. $\frac{-14Gm}{r}$

B. $\frac{-16Gm}{r}$

C. $\frac{-12Gm}{r}$

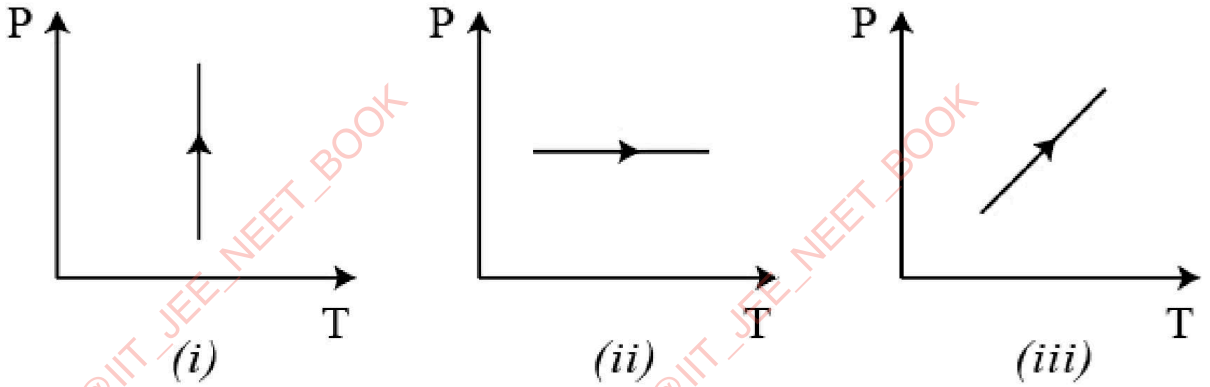
D. $\frac{-8Gm}{r}$

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Pressure versus temperature graphs of an ideal gas are as shown in figure. Choose the wrong statement



A. density of gas is increasing in graph (i)

B. density of gas is decreasing in graph (ii)

C. density of gas is constant in graph (iii)

D. none of these

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If the mean free path of gaseous molecules is 60 cm at a pressure of 1×10^{-4} mm Hg , what will be its mean free-path when the pressure is increased by 100 times ? (Assuming temperature is constant)

A. 6.0×10^{-1} cm

B. 6.0 cm

C. 6.0×10^{1} cm

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D. 6.0×10^2 cm

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A hammer of mass 200 kg strikes a steel block of mass 200 g with a velocity 8 ms^{-1} . If 23% of the energy is utilized to heat the steel block, the rise in temperature of the block is (specific heat capacity of steel = 460 J kg^{-1} K^{-1}) .....

A. 8 K

B. 16 K

C. 12 K

D. 24 K

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A particle is executing simple harmonic motion with an amplitude of 2 m. The difference in the magnitudes of its maximum acceleration and maximum velocity is 4 . The time-period of its oscillation and its velocity when it is 1 m away from the mean position are respectively?

A. 2 s, $2\sqrt{3}$ ms^{-1}

B. $\frac{7}{22}$ s, $4\sqrt{3}$ ms^{-1}

C. $\frac{22}{7}$ s, $2\sqrt{3}$ ms^{-1}

D. $\frac{44}{7}$ s, $4\sqrt{3}$ ms^{-1}

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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A large number of positive charges each of magnitude q are placed along the X-axis at the origin and at every 1 cm distance in both the directions. The electric flux through a spherical surface of radius 2.5 cm centred at the origin is---

A. $\frac{5q}{\varepsilon_0}$

B. $\frac{8q}{\varepsilon_0}$

C. $\frac{3q}{\varepsilon_0}$

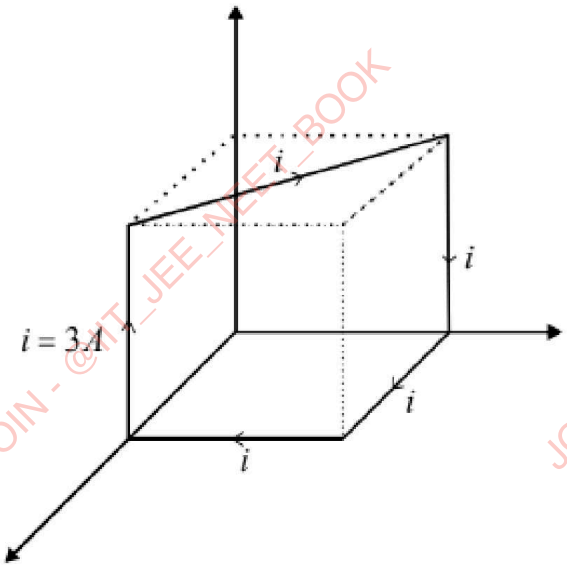
D. ∞

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A wire carrying 3 A current is wrapped around a non - conducting cube of side 1 m as shown below. The magnitude of net magnetic moment due to the loop (in S.I. unit) is M, find 4M.



A. 27

B. 24

C. 18

D. 16

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Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a RC circuit, where R is resistance and C is capacitance which of the following has the dimension of time.

A. $\frac{R}{C}$

B. $\frac{C}{R}$

C. $\sqrt{RC}$

D. RC

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An EM waves enters from medium-(1) to medium-(2). The electric fields are $\vec{E}_1 = E_{01}\sin(kx-\omega t)\hat{j}$ in medium-(1) & $\vec{E}_2 = E_{02}\sin(3kx-\omega t)\hat{j}$ in medium-(2). If $\frac{\mu_{r1}}{\mu_{r2}} = \frac{1}{4}$ then find value of $\frac{4\epsilon_{r2}}{\epsilon_{r1}}$. ($\mu_r$ & ϵ_r are relative permeability & permittivity of medium)

A. 7

B. 8

C. 6

D. 9

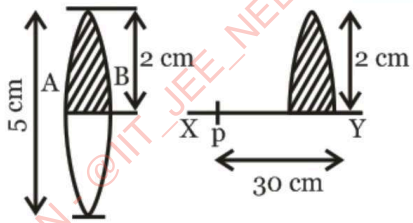
Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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A converging lens of focal length 20 cm and diameter 5 cm is cut along the line AB. The part of the lens shown shaded in the diagram is now used to form an image of a point P placed 30 cm away from it on the line XY. Which is perpendicular to the plane of the lens. The image of P will be formed at a distance _____ from line XY.



A. 0.5 cm

B. 1.5 cm

C. 1 cm

D. 4 cm

Question No. **16**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion: Our ears cannot distinguish two notes, one produced by a violin and other by a sitar, if they have exactly the same intensity and same frequency.

Reason: When a musical instrument is played, it produces a fundamental note which is accompanied by number of overtones called harmonics.

A. Assertion is true, reason is true; reason is a correct explanation for assertion.

B. Assertion is true, reason is true; reason is not a correct explanation for assertion.

C. Assertion is true, reason is false.

D. Assertion is false but, reason is true.

Question No. **17**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Hydrogen atoms absorbs radiation of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which two wavelengths are shorter than λ_0.
1. The final excited state of the atoms is $n = 4$
2. The initial state of the atoms may be $n = 2$
3. The initial state of the atoms may be $n = 3$
4. There are three transitions belonging to Lyman series.

Which of the above are correct?

A. 1, 2, 3, 4

B. 1, 4

C. 2, 4

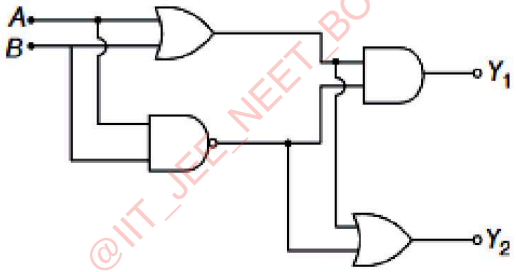
D. 1, 2, 4

Question No. **18**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of Y_1 and Y_2, respectively in the following logic circuit if both A and B are 1.



A. 1,1

B. 1,0

C. 0,1

D. 0,0

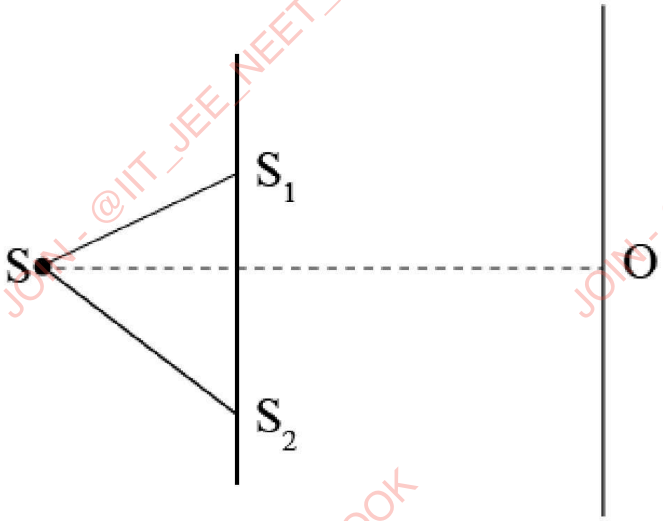
Question No. **19**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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In the setup shown in figure, the two slits, S_1, and S_2 are not equidistant from the slit S. The central fringe at O is, then



A. always bright

B. always dark

C. Either dark or bright depending on the position of S

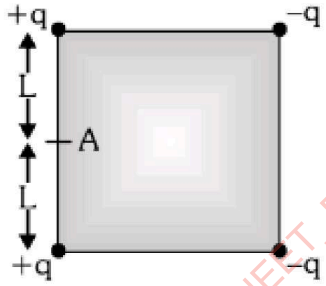
D. Neither dark nor bright

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Four electric charges $+q, +q, -q$, and $-q$ are placed at the corners of a square of side 2 L (Figure). The electric potential at point A, midway between the two charges $+q$ and $+q$, is



A. zero

B. $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}(1+\sqrt{3})$

C. $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1+\frac{1}{\sqrt{5}}\right)$

D. $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1-\frac{1}{\sqrt{5}}\right)$

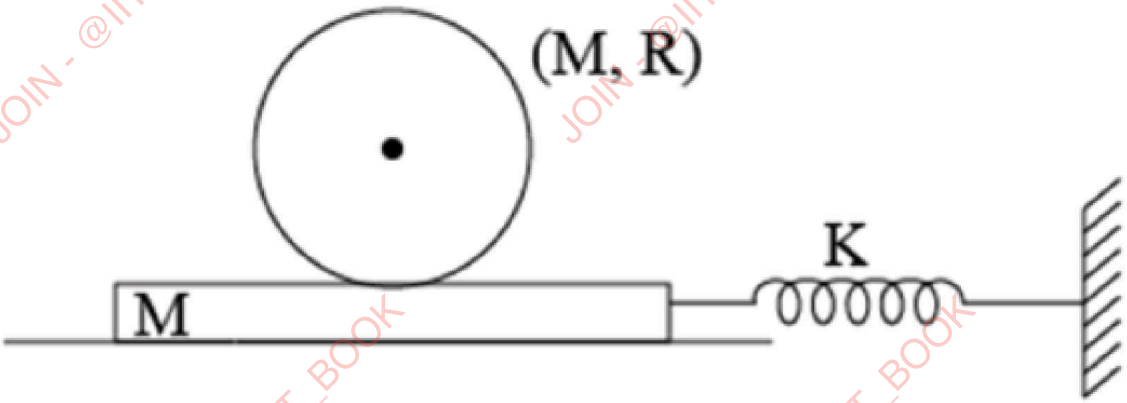
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

A cylinder of mass $M = 2$ kg and radius $R = 12$ cm lies on a plank of the same mass as shown in the figure. The surface between plank and ground is smooth but there is friction between cylinder and plank. If the coefficient of friction between the cylinder and the plank is $\mu = 0.4$, then what maximum initial compression (in cm) can be given to the spring such that the cylinder moves without slipping with respect to the plank? [Given, $k = 200$ N m^{-1}]



Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

A particular force F applied on a wire increases its length by 2×10^{-3} m . If we were to increase the wire's length by 4×10^{-3} m , then the applied force is nF . What is the value of n?

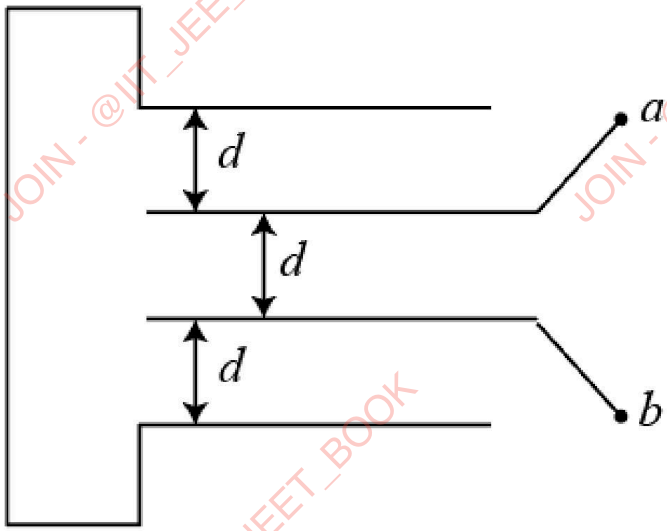
Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Four metal plates, each with surface area A on one side, are placed with separation ' d ' as shown in the figure. The capacitance between a and b is

(ε_0 − permittivity of free space) $\frac{NA\varepsilon_0}{2d}$, Find N.

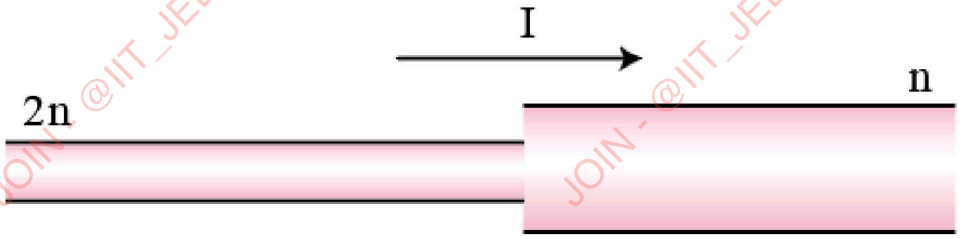


Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Two cylindrical rods of uniform cross-section area A and $2A$, having free electrons per unit volume $2n$ and n respectively are joined in series. A current I flows through them in steady state. Then the ratio of drift velocity of free electron in left rod to drift velocity of electron in the right rod is $(\frac{v_L}{V_R})$ is :



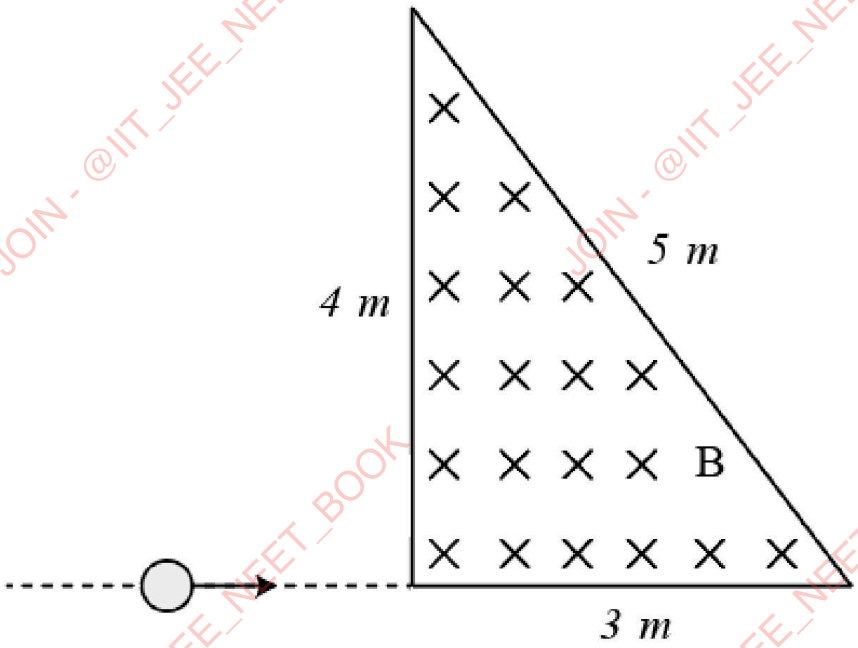
Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

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A small particle of mass 1 kg and charge 1C enters perpendicularly in a triangular region of a uniform magnetic field of strength 2 T as shown in the figure. What is the maximum velocity (in ms^{-1}) of the particle with which it can enter so that it completes a half-circle in the region of the magnetic field?



Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following change decrease the vapour pressure of water kept in a sealed vessel?

A. Decreasing the quantity of water

B. Adding salt to water

C. Decreasing the volume of the vessel to one-half

D. Keeping the temperature of water constant

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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Which of the following order is correct for increasing p-character in orbital used for bonding by central atom---

A. $SiH_4 > CH_4$

B. $H_2S > H_2O$

C. $PH_4^+ > PH_3$

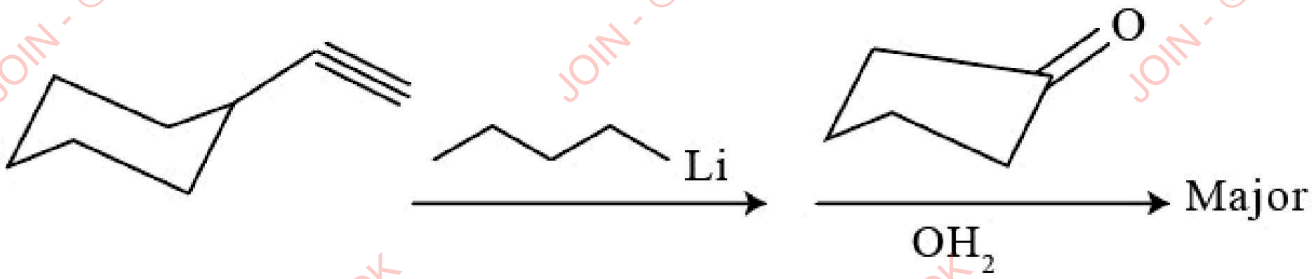
D. $NH_3 > PH_3$

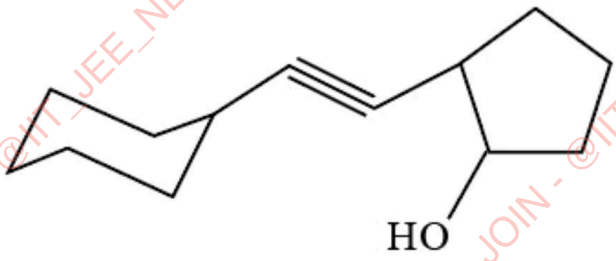
Question No. 3

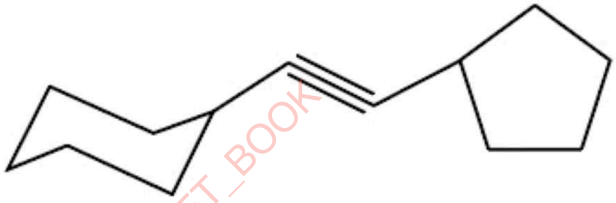
Only One Option Correct Type

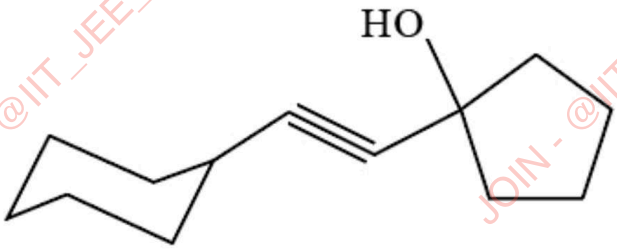
Each question has multiple options out of which ONLY ONE is correct.

Give the major product of the following reaction

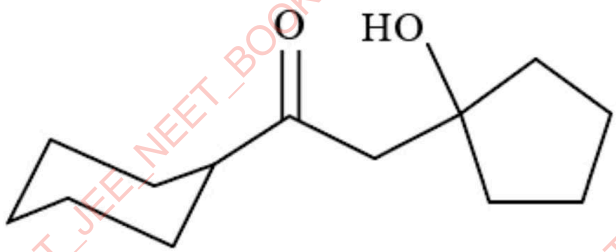


A. 

B. 

C. 

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D. 

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Among the common mineral acids, H_2SO_4 is less volatile due to---

A. hydrogen bonding

B. van der Waals' forces

C. disulphide linkages

D. strong bonds

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The species Ar, K^+ and Ca^{2+} contain the same number of electrons. In which order do their radii increase?

A. $Ar < K^+ < Ca^{2+}$

B. $Ca^{2+} < K^+ < Ar$

C. $K^+ < Ar < Ca^{2+}$

D. $Ca^{2+} < Ar < K^+$

Question No. 6

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Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

CORRECT statement regarding orthoboric acid is :

A. It is a Lewis acid whose pKa is -1.

B. It is a triprotic, weak Bronsted acid.

C. There exist H-bonds in between layers in its crystalline form.

D. Cis-diols can be used to increase acidic strength in its aq. solution.

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Identify the correct statements:

a) In an atom, the possible maximum number of electrons with $n = 4$ and $m_s = +\frac{1}{2}$ is 16.

b) There are 4 sub shells associated with $n = 5$

c) $n = 2, l = 1, m_l = 0$ and $m_s = -\frac{1}{2}$ is a possible set of quantum numbers

d) The number of radial nodes for 3 s orbital is 2

A. a, b, c

B. a, c

C. a, c, d

D. a, b, d

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

$E^{\ominus}$ values for the couples Cr^{3+}/Cr^{2+} and Mn^{3+}/Mn^{2+} are -0.41 and $+1.51$ volts, respectively. Considering these values select the correct option from the following statements.

Cr^{2+} acts as a reducing agent and
Mn^{3+} acts as an oxidising agent in their
A. aqueous solutions.

B. Cr^{2+} (aq) is more stable than Cr^{3+} (aq).

Mn^{3+} (aq) is more stable than
C. Mn^{2+} (aq).

D. None of these.

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following reactions are not feasible?

I. $CH_3CH=CH_2 \xrightarrow[(C_6H_5CO)_2O_2]{HCl} CH_3CH_2CH_2Cl$

II. $C_6H_6 + CH_3CH_2CH_2Cl \xrightarrow{AlCl_3} C_6H_5CH(CH_3)_2$

III. $CH_3CHClCH_2Cl \xrightarrow[\Delta]{KOH} CH_3C \equiv \overset{+}{C}H$

IV. $CH_3CH=CHCH_3 \xrightarrow{KMnO_4/H^+} CH_3COOH$

A. I & IV only

B. II & III only

C. II & IV only

D. I & III only

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following ions has the highest magnetic moment---

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A. Ti^{3+}

B. Sc^{3+}

C. Mn^{2+}

D. Zn^{2+}

Question No. **11**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the following List - I with List - II

List - I	List - II
P) $Na_2[Zn(OH)_4] + H_2S \rightarrow$	1) Precipitate is formed through non - redox reaction
Q) $Hg\downarrow + HgS\downarrow + \underset{(excess)}{aqua\text{ - }regia} \xrightarrow{\Delta}$	2) No precipitate is formed but a redox reaction occurs
R) $Na_2S_2O_3 + KI_3(Aq.) \rightarrow$	3) A violet colouration formed as intermediate
S) $Fe^{3+} + S_2O_3^{2-} \rightarrow$	4) neutral gas is evolved

Codes :

A.

P	Q	R	S
4	1	3	2

B.

P	Q	R	S
4	1	2	3

C.

P	Q	R	S
1	2	3	4

D.

P	Q	R	S
1	4	2	3

Question No. **12**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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The correct order of boiling points of H_2O, H_2S, H_2Se and H_2Te respectively is----

A. $H_2O > H_2S = H_2Se = H_2Te$

B. $H_2O < H_2S < H_2Se < H_2Te$

C. $H_2O > H_2S > H_2Se > H_2Te$

D. $H_2O > H_2Te > H_2Se > H_2S$

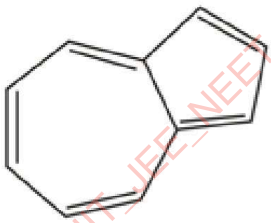
Question No. **13**

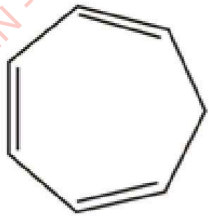
Only One Option Correct Type

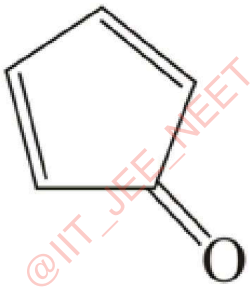
Each question has multiple options out of which ONLY ONE is correct.

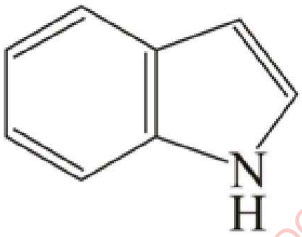
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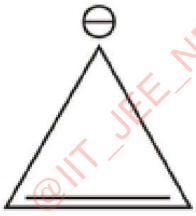
Among the following.

I 

II 

III 

IV 

V 

The antiaromatic compounds are

A. I and IV

B. III and V

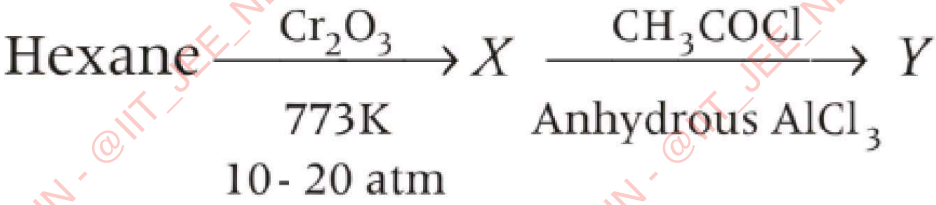
C. II and V

D. I and III

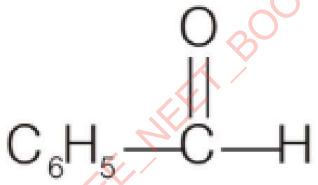
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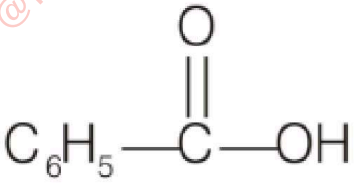
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

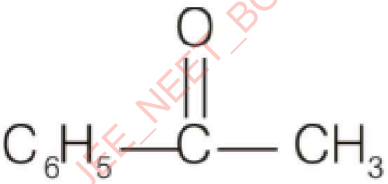


Y in the above sequence of reaction is

A. 

B. 

C. $C_6H_5 - CH_3$

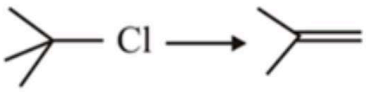
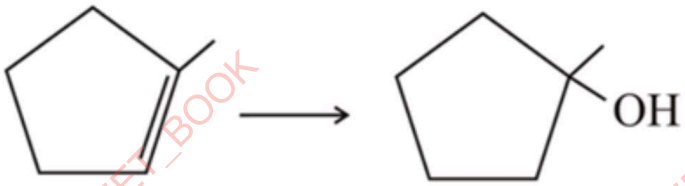
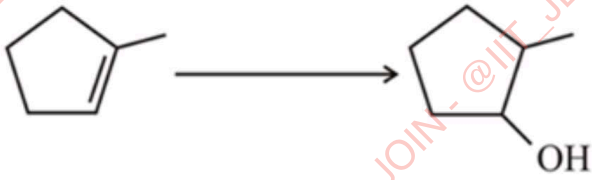
D. 

Question No. **15**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the chemical conversions in list I with appropriate reagents in list II and select the correct answer using the code given below the list.

List I		List II	
P.		I.	(i) $Hg(OAc)_2$; (ii) $NaBH_4$
Q.		II.	NaOEt
R.		III.	Et – Br
S.		IV.	(i) BH_3; (ii) $H_2O_2/NaOH$

A. P – II, Q – III, R – I, S – IV

B. P – III, Q – II, R – I, S – IV

C. P – II, Q – III, R – IV, S – I

D. P – III, Q – II, R – IV, S – I

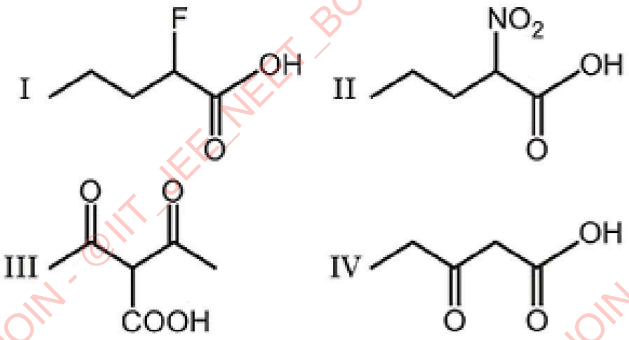
Question No. **16**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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Arrange the following carboxylic acids in increasing order of reactivity towards thermal decarboxylation reaction



A. I < II < III < IV

B. I < III < IV < II

C. I < IV < II < III

D. IV < I < II < III

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the titration of I_2 (aq) by $S_2O_3^{2-}$ (aq) using the starch indicator, the end point is indicated by:

A. Colourless to blue

B. Blue to colourless

C. Pink to colourless

D. Blue to pink

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Sucrose is a disaccharide and a non-reducing sugar.
Reason (R): Sucrose involves glycosidic linkage between C_1 of β-glucose and C_2 of α-fructose.
Choose the most appropriate answer from the options given below :

A. Both (A) and (R) are true and (R) is the true explanation of (A).

B. (A) is true but (R) is false.

C. (A) is false but (R) is true.

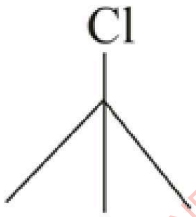
D. Both (A) and (R) are true but (R) is not the true explanation of (A).

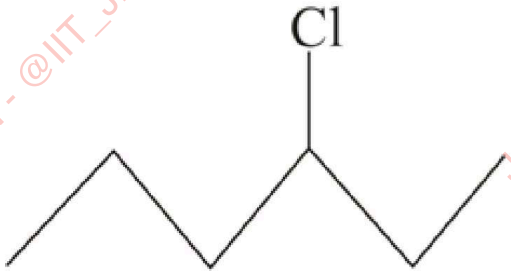
Question No. **19**

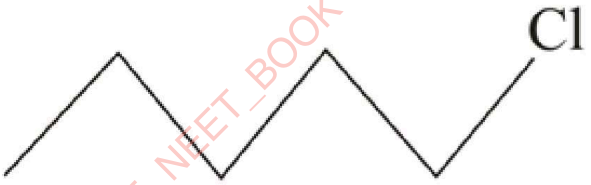
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

What is the order of reactivity of following compounds towards S_N1 substitutions reaction.

(i) 

(ii) 

(iii) 

A. (ii) > (i) > (iii)

B. (iii) > (ii) > (i)

C. (i) > (ii) > (iii)

D. None of these

Question No. 20

Only One Option Correct Type
Each question has multiple options out of which ONLY ONE is correct.

The qualitative distinction of $ZnSO_4$ and $Al_2(SO_4)_3$ can be done by using the reagent :

A. NH_4OH

B. NaOH

C. Any of these

D. none of these

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type
The answer has to be filled into the input box provided below.

For the reaction $3A \rightleftharpoons C + 2D$, initially A was taken. At equilibrium the concentration of D is twice that of A. The value of K_c is

Question No. 2

Numerical Type
The answer has to be filled into the input box provided below.

One mole of an ideal monoatomic gas is heated in a process $PV^{5/2} =$ constant. Amount heat absorbed in the process for $36^\circ C$ rise in temperature. (Answer in Cal)

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Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

In an experiment to estimate the molecular weight of benzoic acid by elevation in boiling point method, the experimental value of molecular weight was double the actual value. Calculate the degree of association of dimer, if the elevation in Boiling Point is 2° C.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

If k is rate constant and A is collision frequency then find $(Ea_2 - Ea_1)$ for reaction in kcal/mol at 227° C

$k_1 = 100k_2$

$10\ A_1 = A_2$

$\ln 10 = \frac{7}{3}$

(Nearest Integer)

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

0.2gm of an organic compound was analysed by Kjeldahl's method, Ammonia evolved was absorbed in 60 mL of N/5 H_2SO_4 . Unused acid required 40ml of N/10 NaOH for complete neutralization. The percentage of nitrogen in the compound is :

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of complex numbers z satisfying the conditions $|(z/\bar{z}) + (\bar{z}/z)| = 1, |z| = 1$ and $\arg z \in (0, 2\pi)$ is

A. 4

B. 12

C. 16

D. 8

Question No. **2**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = [a_{ij}]$ be a square matrix of order 3 and $B = [b_{ij}]$ be a matrix such that $b_{ij} = 2^{i-j}\, a_{ij}$ for $1 \le i, j \le 3,\ \forall i, j \in N$. If the determinant of A is same as its order, then the value of $\left|(B^T)^{-1}\right|$ is

A. $\frac{1}{3}$

B. 3

C. 9

D. $\frac{1}{27}$

Question No. **3**

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The mirror image of the curve $\arg(\frac{z-3}{z-i}) = \frac{\pi}{6}$ in the real axis is

A. $\arg(\frac{z+3}{z+i}) = \frac{\pi}{6}$

B. $\arg(\frac{z-3}{z+i}) = \frac{\pi}{6}$

C. $\arg(\frac{z+i}{z+3}) = \frac{\pi}{6}$

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D. $\arg(\frac{z+i}{z-3}) = \frac{\pi}{6}$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = x A^2 + yA + zI$ then find the value of $(x + y + z)$ where I is a unit matrix of order 3.

A. 2

B. 5

C. 1

D. 3

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a, b, c are non zero real numbers, then minimum value of the expression $(\frac{(a^4+a^2+1)(b^4+7b^2+1)(c^4+11c^2+1)}{a^2b^2c^2})$, , is -

A. 315

B. 351

C. 415

D. 451

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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Let the sets $A = \{2, 4, 6, 8 \ldots\}$ and $B = \{3, 6, 9, 12 \ldots\}$ such that $n(A) = 200$ and $n(B) = 250$. If $n(A \cup B) = k$, then k is equal to

A. 464

B. 384

C. 450

D. 396

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = (x-1)^2(x-2)^3 e^x$. Then

A. f(x) has local maximum at x = 1

B. f(x) has point of inflexion at x = 1

C. f(x) has local minimum at x = 2

D. all of these

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $F(x) = \int_{x}^{\frac{\pi}{3}+x^2} 2\cos^2 t \, dt$, $x \in R$ and $f: [0, \frac{1}{2}] \to [0, \infty)$ be a continuous function. For $a \in [0, \frac{1}{2}]$. If $F'(a) + 2$ is the area of the region bounded by $y = f(x)$, $x = 0$, x -axis and $x = a$, then find $f(0)$.

A. 2

B. 0

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C. 3

D. 1

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The value of $I = \int_{-1}^{1} [x\sin(\pi x)]dx$ is (where $[\cdot]$ denotes the greatest integer function)

A. π

B. 2π

C. 0

D. $-\pi$

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The shortest distance between the skew-lines $\mathbf{r} = (-\hat{\mathbf{i}} + 3\hat{\mathbf{k}}) + t(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$ and $\mathbf{r} = (3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) + s(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ is

A. $\frac{10}{\sqrt{17}}$

B. $\frac{22}{\sqrt{17}}$

C. 9

D. 8

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

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The solution of the differential equation $\frac{dy}{dx} = \frac{y^2 + x \ln x}{2xy}$ is (where, c is the constant of integration)

A. $2x^2 = y(\ln x)^2 + 2cy$

B. $2y^2 = x(\ln x)^2 + 2cx$

C. $x^2 = y(\ln x)^2 + c$

D. $2y^2 = \frac{x}{y}(\ln x)^2 + cx$

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For a parabola passing through $(1, 2), (2, 1), (3, 4)$ and $(4, 3)$, the equation of axis of the parabola is-

A. $x - y + 3 = 0$

B. $x - y = 0$

C. $x + y - 1 = 0$

D. line $x + y = 0$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If point $P(\cos\theta, \cos\theta)$ lies outside the circle $x^2 + y^2 + 2x + y - 2 = 0$, then the total number of values of n which exist if $\theta = \frac{n\pi}{6}$, $n \in N$ and $\theta \in [0, 2\pi]$, are -

A. 0

B. 3

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C. 5

D. 7

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If Q is the foot of the perpendicular from the point $P(4, -5, 3)$ on the line $\frac{x-5}{3} = \frac{y+2}{-4} = \frac{z-6}{5}$ then $25(PQ)^2 - 450 =$

A. 9

B. 8

C. 6

D. 7

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a parallelogram OACB, $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$ & foot of perpendicular drawn from point B to AC is M. If $\vec{a} \cdot \vec{b} = 1$ & $|\vec{a}| = |\vec{b}| = 2$, then $|\overrightarrow{BM}|$ is $\sqrt{K}/2$, find K.

A. 15

B. 14

C. 16

D. 13

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

For the set $A = \{x_1, x_2, x_3, x_4, x_5\}$ the variance is 4 and the mean is 2. For the set $B = \{y_1, y_2, y_3, y_4, y_5\}$ the variance is 5 and the mean is 4. Then, the variance of $A \cup B$ is

A. 6

B. 6.5

C. 5.5

D. 5

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Ravish and Vineet decided to meet on New Year's day evening at a place in Cannaught Place between 7pm and 8pm. The first one to arrive waits for 20minutes and then leave. Assuming that each is independently equally likely to arrive at any time during the hour (between 7pm and 8pm), then the probability that they meet is

A. $\frac{5}{16}$

B. $\frac{3}{16}$

C. $\frac{3}{8}$

D. $\frac{5}{9}$

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $y = \tan^{-1}\frac{1}{1+x+x^2} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \ldots +$ upto $2n$ terms $(\forall x \geq 0)$, then $y(0)$ is

A. $\tan^{-1}(n)$

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B. $\tan^{-1}(2n)$

C. $2\tan^{-1}(n)$

D. 0

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The quadratic equation whose roots are $\sin^2 18^\circ$ and $\cos^2 36^\circ$ is :

A. $16x^2 - 12x + 1 = 0$

B. $16x^2 + 12x + 1 = 0$

C. $16x^2 - 12x - 1 = 0$

D. $16x^2 + 10x + 1 = 0$

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find number of integral values of λ if $(\lambda, \lambda + 1)$ is an interior points of $\triangle ABC$, where $A \equiv (0, 3), B \equiv (-2, 0)$ and $C \equiv (6, 1)$.

A. 5

B. 1

C. 3

D. 2

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

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Numerical Type

The answer has to be filled into the input box provided below.

If the number of integral values of x satisfying the inequality $\frac{(e^x-1)(\sin x-2)(x^2-5x+4)}{x^2(-x^2+x-2)(2x+3)} \leq 0$ is λ, then λ^2 is equal to

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

If A is an invertible matrix of order 3 and B is another matrix of the same order as of A, such that $|B| = 2$, $A^T|A|B = A|B|B^T$. If $\left|AB^{-1}\ \text{adj}(A^T B)^{-1}\right| = K$, then the value of 16K is equal to

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

Area bounded by the curves $y = e^{x^2}$, the x-axis and the lines $x = 1, x = 2$ is given to be α square units. If the area bounded by the curve $y = \sqrt{\ell nx}$, the x-axis and the lines $x = e$ & $x = e^4$ is expressed as $(pe^4 - qe - \alpha)$, (where p and q are positive integers), then $(p + q)$ is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let $f[1, \infty) \to [2, \infty)$ be a differentiable function such that $f(1) = \frac{1}{3}$. If $6\int_1^x f(t)dt = 3xf(x) - x^3$ for all $x \geq 1$, then the value of $3f(2)$ is

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

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If 7^{103} is divided by 25 , then the remainder is

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